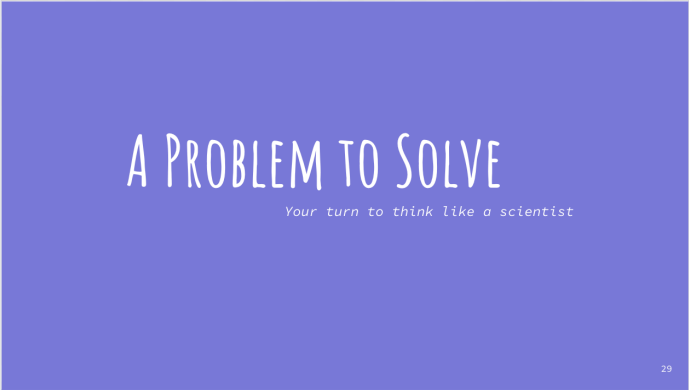


Reasoning:

This lesson provides several opportunities for students to use their own experiences with Covid-19 to examine a phenomenon that is new or emergent. By elevating individual student experiences with Covid-19, they will be more motivated to arrive at a meaningful conclusion for the simulated town and transfer the process to their lives outside the classroom. While exploring the simulation students are introduced to a problem of inquiry; they must rely on the thinking of themselves and their peers to make sense of a new situation. The students are also given an opportunity to use the class criteria for good scientific practices to analyze experimental designs using the simulation. This lesson provides students with the mental schema required for the subsequent lessons focused on methodology design, data collection, and conclusion generation.

Slide	Notes
DO NOW! LESSON 3 Do you remember your experiences during the height of the COVID epidemic? We were all given advice on how to stay safe. Do you remember what you did to stay safe? How did you decide what to do? Instructions: Turn and talk to a partner about your thoughts, and type them in our "Thoughts & Reflections" sheet (in the tab with your name on it).	Students can discuss in pairs or as a class   Prompt students to draw from their experience of relevant infectious diseases to get them thinking about how scientists design experiments to evaluate phenomena that are new to them, as well as how scientific information is communicated to others.
	Inform students that in this unit, they will take the role of scientists to solve a problem that has emerged which is putting lives at risk.   It is important to highlight for students at this early stage in the activity: 1. They will be addressing a large-scale problem the same way scientists approach problems. 2. The conclusions they eventually agree upon will be determined through the sound use of scientific practices. Either read out the next slides yourself, have a student read out the next slides to the whole class, or have them read in groups and debrief the scenario with students.

One more thing to consider:

As scientists, it's crucial to recognize that scientific decisions impact people. Sometimes people make decisions that do not follow the scientific reasons. For example, during COVID people were asked to social distance from each other but there were some jobs where that was impossible to do, such as nurses and doctors. So there are trade-offs to consider when scientific knowledge is applied in the real world.

Use this slide to highlight the need to study the problem in a manner that humanizes the implications that may emerge from their investigations. 🔍

Use this as an opportunity to signal that every decision involves tradeoffs and it is imperative for scientists to make evidence-based suggestions and be transparent about possible limitations of the suggestions they make.

What are your best recommendations of mitigation strategies for the public to stop the spread of this disease and how did you arrive at your conclusions?

Your Research Question

Possible check for understanding questions: 💡

- What do you think the problem is?
- What is required of the scientists?
- Does this problem seem familiar to you?
- What does “mitigation” mean in this context?

This helps students frame the research question so they are all on the same page. This will be helpful later when exploring the value of meta-analysis and drawing conclusions from data that seeks to answer the same question.

LET'S EXPLORE OUR TOOL
- THE SIMULATION

Introducing students to the simulation. This can be either teacher-guided, student-initiated or a combination of both. 🔍

- The teacher may choose to walk students through the slides provided or send them to explore in their groups following the protocol on Slide 47. Teachers can also employ both strategies if time allows).
- The description of each slide can be found in the speaker notes. Either the teacher or students can read through these notes.

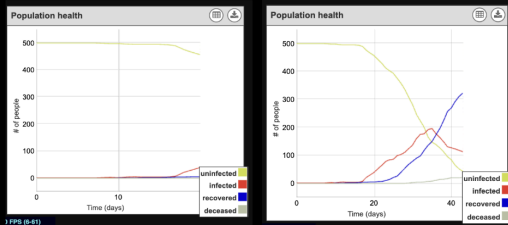
The disease specifications of this model is roughly based on H1N1 (swine flu). The students should not know this as they explore the simulation and develop their own conclusions. Lesson 7 reveals this to the students, though it could also be revealed at the end of Lesson 6 if Lesson 7 is skipped.

SOME POSSIBLE DISCUSSION POINTS:

1. Does this model include all of the parts you would expect based on your experience?
2. If not, do you think the model has limitations?
3. How can you use this model to address your specific research question?
 - These additional questions can help students evaluate the model with a more discerning eye and think about

effective ways to test their specific question. This can help lead fluidly into Lesson 4 when they will plan their experiment.

NOTICE THE 10 DAYS VS 40 DAYS TIME INTERVALS



The simulation shows results at varying time intervals. For example, the graphs show the trend of handwashing across 0, 10, 20, and 40.

Ask students to look at the graphs and elicit differences they notice in the pattern of yellow (uninfected), infected (red), recovered (blue) and grey (deceased) lines as the time interval increases. Use the discussion to tell students time intervals therefore become an important variable to make note of when planning and running experiments. 💡

Important ideas to point out: 🔬

- Accounting for variables such as time intervals, help us notice the various conditions under which we got the results we did.
- This information in turn, helps other scientists replicate our findings, which if they do, can increase the reliability of the claims we make.

WHAT SHOULD YOU BE DOING WITH YOUR GROUP? - 10 MINS

1. Use the following features on the simulation:
 - a. Setup, run, edit camera, various mitigation strategies (for example, handwashing, vaccines), engine speed, table view
2. Try out different mitigation strategies:
 - a. Choose to enable one mitigation strategy, run the simulation, and observe the trend
 - b. Try collecting data from one trial and compare it to what you get by averaging the results of three trials

Have students play with the simulation model and record observations in any way they choose. 💡 🔍

Questions to consider asking students while they are working:

- Is there any benefit from doing multiple trials instead of one trial?
 - Possible answer: Averaging results of multiple trials help us systematically evaluate evidence as it helps account for random variation and minimizes biases/ errors that can affect a single trial.

EVALUATING PLANS OF INQUIRY

Ask students if they have ideas on how they want to set up their experiments to determine the effectiveness of a single mitigation factor 💡

- Some leading questions include:
 - Do you have one mitigation factor you are curious to investigate?
 - How would you go about investigating its effectiveness?

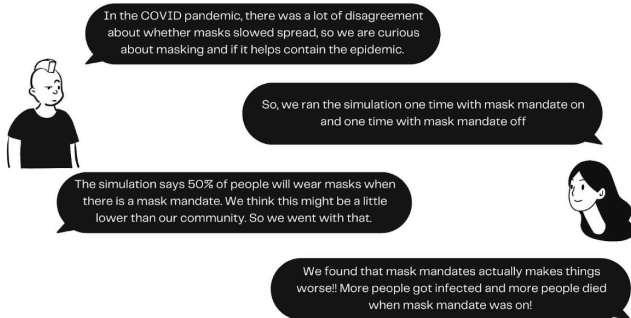
Use this discussion as an opportunity to guide their attention to reference norms from the 'criteria for good experiments' to determine quality of experiments. 🔍

Tell students we are going to take a break from playing with the simulation and now evaluate some

experimental designs students in the past have come up with to evaluate mitigation factors. The class will go through these experimental designs and evaluate their quality using the class criteria. 💡🔍
They will start by

- Reading the rationale of students' experiment (this is WHAT the students decided to examine and WHY)
- Then, they will evaluate the experimental designs put together by the group (the HOW the students decided to examine the mitigation factor of interest)

Have 2 students read out the rationale of these two students.



Experiment Introduction

We want to find out if mask mandates reduce the number of infected and deceased people. We ran simulations with and without mask mandates.

Condition 1 Method		
Mitigation Factors	[on/off]	# of Trials
Handwashing	off	1
Vaccines	off	1
Lockdown	off	# of Days
Virtual Work/School	off	40
Mask Mandates	on	40

Condition 1 Results	
Results	Trial 1
Infected	63
Recovered	361
Deceased	11

Condition 2 Method		
Mitigation Factors	[on/off]	# of Trials
Handwashing	off	1
Vaccines	on	1
Lockdown	off	# of Days
Virtual Work/School	off	40
Mask Mandates	off	40

Condition 2 Results	
Results	Trial 1
Infected	100
Recovered	301
Deceased	3

Conclusions from Experiment

Mask mandates actually make the epidemic worse. Mask mandates increased the total number of people that got infected by 23 (63 vs. 40) and the number of deceased people by 8. So mask mandates do not work.

EVALUATE THEIR STUDY

Ask students to critique this experiment using their criteria for good methods. 🔍

There are three problems that they can notice, and all of these makes the final inference problematic.

1. The experiment is confounded. The "control" condition has mask mandates and virtual/school work on, so this is not a fair comparison. This is not comparing lockdown on with lockdown off, with all other variables controlled.
2. There is only 1 trial per condition, which is too few.
3. Students might also notice that 40 days is OK, but sometimes not enough for many of the infections and deaths to occur in the simulation. (40 days often gives results similar to a longer number of days, though.)

Have students discuss how to fix this design. The most important fixes are to increase the number of trials per condition and to get rid of the confound. One way to fix the confound is to turn the vaccinations off in the control condition. Then you're finding out the effects of mask mandates all by themselves, without any other mitigation strategies.



- You can't conclude that taking away mask mandates made things better, when you also

added vaccinations at the same time. The so-called “control” condition might have been better because the students turned vaccinations on, and now people were getting vaccinated.

- *Teaching note:* it is also possible to turn the vaccinations ON in both conditions, as long as it’s the same in both conditions. Help students see that when researchers make this choice, they’re finding out something else that is interesting--whether mask mandates add to protection if vaccinations are already available. What the control conditions are depends on what you want to know. This idea will help prepare them for the discussions about disagreements coming up later.

Emphasize that students can use their class list of characteristics of good methods to help evaluate this method, and to create good methods themselves.

- You can ask students at the end if there are any modifications/revisions they would like to make to their list of characteristics of good methods.
- You can also ask students how these critiques improve their scientific knowledge (orienting to epistemic practices), and how this relates to what scientists do (orienting to the discipline).
- Emphasize that we want to fit our conclusions to the high-quality evidence, and it’s appropriate to dismiss studies that are flawed.

END OF LESSON 3 REFLECTION

Based on your experience in class today, how would you decide what to do to stay safe in a future epidemic?

Instructions:
Write your thoughts on this in our “Thoughts & Reflections” sheet (in the tab with your name on it).

Elicit experiences about COVID-19 to get students to reflect about the nature of scientific investigations when scientists are tasked with studying a new or an unfamiliar phenomenon. 💡🌐🔬🔍

- Is it bad when scientists do not find support for their initial predictions or if they change their mind about how to investigate a phenomena?

At the end of this lesson student reflections should include elements of reliable science practices and the impact they have on conclusions that are made or solutions to problems.